

Certified analytical bounds for hypersonic reachable sets

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Abstract

Computing exact reachable sets for six-degree-of-freedom hypersonic entry vehicles is intractable for grid-based dynamic programming and compromised by the floating-point approximations of numerical solvers. This research introduces a purely analytical framework to compute mathematically certified bounds using exact interval arithmetic and occupation measures. By modeling transcendental aerodynamics natively as interval differential inclusions, the method circumvents artificial state augmentation and numerical optimization entirely. Utilizing a thin-atmosphere limit profile decomposition to separate the equilibrium-glide skeleton from compact skip events, the framework yields explicit, closed-form algebraic bounds that provide a strict mathematical certificate in the minimal physical state space.

1 The problem

The reachable set of an entry vehicle (every state it can be driven to under admissible control) is what safety and threat analyses need bounded. Computing it exactly means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states [1, 2]. A rigid-body entry vehicle carries thirteen states,

$$x = (r, \lambda, \phi, V, \gamma, \psi, q_0, q_1, q_2, q_3, p, q_b, r_b) \in \mathbb{R}^{13}, \quad (1.1)$$

and $17 + 2N_m$ once ablation recession, integrated heat load and aerothermoelastic deformation are coupled [3]. The translational state is carried in geocentric spherical position (r, λ, ϕ) and wind-frame velocity (V, γ, ψ) so that the entry-corridor constraints (bounded altitude, bounded speed, heating and load-factor limits) are box constraints on states, the natural setting for interval arithmetic. Attitude is a unit quaternion, free of Euler-angle singularities.

So the exact grid-HJI method quits before the real problem starts.

The standard polynomial barrier with coefficients obtained from an interior-point solver has analytical *form* but inherits the solver’s accuracy [4], so it is not a certificate; a numerical solver’s failure modes are documented, and they include the loss of strict feasibility on which every rounding-based scheme depends [5, 6]. There are then two problems to solve, not one: dimension, and the provenance of the guarantee. The second is not a matter of care. Interval arithmetic is the setting in which a guarantee has meant, for forty years, an enclosure rather than an approximation, and the property that makes an enclosure valid at all—inclusion isotonicity—is exactly what a reimplement loses first; not every law of real arithmetic survives the move to intervals, which is the original form of the wrapping effect [7, 8]. Applied interval analysis supplies the machinery the pipeline would be built from, and it is deliberately conservative: certified set inversion, deterministic intersection geometries, and a customized exact-arithmetic integrator to ensure floating-point inaccuracies do not compromise the final bounds [9].

The proposal demands a guarantee verified in exact arithmetic rather than produced by a numerical solver. Floating-point inaccuracies inherent to standard simulation tools compromise the analytical certificate. Must implement a custom, exact-arithmetic interval integrator (likely leveraging C/C++ or specialized rational arithmetic libraries). This pipeline must process the hybrid dynamics, manage the boundary-layer bubble transitions, and compute the intersection geometries deterministically to verify the final bounds.

2 The approach

Embed the hybrid flight dynamics into a Banach space of occupation measures. The nonlinear flight mechanics $\dot{x} = f_q(x, c, t)$ become *linear*: the Liouville equation is a linear constraint on a measure μ over $[0, T] \times \mathcal{X} \times \mathcal{U}$, and $\text{supp } \mu_T$ outer-approximates the reachable set. So far this is the occupation-measure program of Lasserre, Henrion and collaborators [10, 11]. The embedding buys linearity at the cost of uniqueness, and the way it is recovered is standard: a trajectory family satisfying the Liouville constraint lies in the convex hull of the generalized curves, so measures correspond to convex mixtures of trajectories rather than to one of them [12, 13]. The same relaxation is what admits the interval envelope, and that is the first thing that has to be built rather than borrowed. Because the true aerodynamics are transcendental, they are enclosed in strict intervals, and a vector field selected from an interval is a set-valued map; the resulting differential

inclusion is the object the measure must describe, not the nominal field [14, 15]. Occupation-measure theory for inclusions of exactly this kind already exists and must be cited as prior art: it defines an occupational invariant measure without assuming solution uniqueness, which is the same move made here [16]. What it assumes is worth stating plainly, because it is where the entry problem is likely to fail: the vector fields must satisfy a linear growth condition, and the singular-perturbation results additionally require a uniform bound on the fast state along the whole ε -family [16, 17].

Objective 2: Liouville equation for differential inclusions

The current framework embeds the flight mechanics $\dot{x} = f_q(x, c, t)$ as a linear Liouville constraint on an occupation measure μ over $[0, T] \times \mathcal{X} \times \mathcal{U}$. By bounding true non-polynomial aerodynamics within strict intervals, the vector field becomes a differential inclusion $\dot{x} \in \mathcal{F}(x, c, t)$. We must formulate a robust Liouville equation capable of acting on set-valued maps, and rigorously prove that the support of the resulting occupation measure, $\text{supp } \mu_T$, securely outer-approximates all possible trajectories generated by any vector field selected from the interval inclusion.

The new content is what happens in the thin-atmosphere limit $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$. Rescaled by the phugoid-to-secular time ratio (which, as in the classical Chapman–Loh variables, enters as $\sqrt{\varepsilon_{\text{atm}}}$, not ε_{atm} [18, 19]), the occupation-measure family *is not tight*. Mass concentrates onto the low-dimensional equilibrium-glide slow manifold $\Lambda = 1$. In the 3-DOF (point-mass) approximation this is the scalar condition $L \cos \sigma / [m(g - V^2/r)] = 1$; in the full 6-DOF formulation the lift component $L \cos \sigma$ is not an independent input but is determined by the attitude quaternion through the body-to-LVLH rotation, so the slow manifold is a single scalar constraint on the 13-dimensional translational-rotational state (thirteen coordinates; twelve degrees of freedom after the unit-quaternion constraint). The full dynamics carry J_2 oblate gravity throughout: the gravitational acceleration has both a radial component $g_D(r, \phi)$ and a northward component $g_N(r, \phi)$, and the general slow-manifold condition retains both. The simplified scalar $\Lambda = 1$ is the leading-order reduction under spherical gravity and small flight-path angle. Retaining J_2 is not a refinement but a necessity: during skip maneuvers the northward component g_N produces latitude-dependent heading drift, the radial component g_D varies the centrifugal balance at re-entry by up to $\approx 0.65\%$ of the leading gravitational term, and these effects compound over multiple skips.

The loss of tightness is read on the *defect measure* of the rescaled family. The occupation measures live in the cone $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$ of nonnegative Radon measures, topologized weakly-* as the dual of $C_0([0, T] \times \mathcal{X} \times \mathcal{U})$. Writing the weak-* limit as $\mu_\varepsilon \rightarrow \mu_0 + \nu$, with μ_0 the limiting glide-manifold measure, the *defect measure* $\nu \geq 0$ is the nonnegative remainder—the mass that fails to concentrate on the slow manifold as $\varepsilon_{\text{atm}} \rightarrow 0$. On ν the situation is the measure-theoretic analogue of Lions’ concentration alternative—vanishing, dichotomy, or concentration [20]. The relevant scaling group is the phugoid time dilation $t \mapsto t/\sqrt{\varepsilon_{\text{atm}}}$ that localizes each atmospheric pass, so the bubbles are concentration at a shrinking timescale and it is the dilation half of the principle—the limit-case modification—that applies, not the translation (escape-to-infinity) case [21]. Rather than restore compactness, we extract the profile decomposition the dichotomy hands us:

- a finite-dimensional *skeleton* S —the concentration component, the glide manifold as a graph over (r, V) in the corridor subspace, for which a rigorous characterization with quantitative accuracy estimates already exists [22];
- one *boundary-layer bubble* per skip event, the analogue of a concentrating profile: a compact impulse set $\mathcal{I}_j$ of dimension $\leq m$, carried by an internal scaling group at each atmospheric pass, with orthogonality read off the scale-and-core separation condition that governs concentrating profiles generally [23]. The J_2 oblate-gravity terms are retained in the bubble dynamics: the

northward component g_N introduces a latitude-dependent heading drift during each skip, and the variation of g_D with latitude modifies the centrifugal balance at re-entry. These effects are cumulative over multiple skips and must be enclosed for tight certified bounds;

Boundary-layer bubbles represent discrete atmospheric skip events as compact impulse sets $\mathcal{I}_j$ carried by internal scaling groups. Because our framework fixes time as the independent variable, integrating interval bounds through highly dynamic skip events will induce severe “wrapping effects” (artificial inflation of bounding boxes). Must develop advanced intersection geometries (zonotopes, Taylor models, etc.) integrated into the profile decomposition to prove geometries maintain tight corridor bounds through multiple skip events without the reachable set exploding prematurely.

- a *compact residual* with a quantitative bound, $O(\sqrt{\varepsilon})$ in the corridor metric, in a norm chosen so that the loss of compactness actually disappears rather than merely being hidden [24].

The number of bubbles is bounded *a priori*, $K \leq K_{\max}(Q_{\max})$, by the integrated heat-load budget (a trajectory can only afford so many skips [25, 26]). The wrapping effect to be suppressed here is not new: for linear systems it is already quantified, and the term is first order in the time step and exponential in the horizon times the matrix norm [27]. The multi-parameter separation of the three scaling groups (atmospheric, attitude, elastic) is the Bahouri–Gérard almost-orthogonality argument, transported to defect measures: two profiles are almost-orthogonal when their scale parameters diverge, so that their rescaled supports become asymptotically disjoint and their cross terms vanish in the weak-* pairing on $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$. Establishing this non-interaction for all three groups simultaneously is the one genuinely new obligation, and it sits on the critical path.

The framework relies on the Bahouri–Gérard almost-orthogonality argument to separate the multi-parameter scaling groups (atmospheric, attitude, and elastic). Under interval aerodynamics, these groups couple with an adversarial, bounded uncertainty. We must prove the non-interaction theorem for the defect measures while accounting for interval bounds, guaranteeing that the gap between the upper and lower aerodynamic bounds does not cause mass to leak between scaling groups during atmospheric transition, preserving the compact residual bound $O(\sqrt{\varepsilon})$.

The argument worth transporting is the constructive one: splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so orthogonality is produced by the construction rather than assumed by it [28].

The target statement is

$$\mathcal{R}_{\text{cert}}(w) \subseteq \underbrace{\left(\underbrace{\Phi_S(\Theta)}_{\text{skeleton}} \oplus \underbrace{\bigoplus_{j=1}^K \mathcal{I}_j}_{\text{bubbles}} \right)}_{\text{residual}} \oplus \underbrace{\mathbb{B}(C_{\mathcal{R}}\sqrt{\varepsilon})}_{\text{residual}}, \quad (2.1)$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. The exact residual corridor constant $C_{\mathcal{R}}$ and the maximum bubble count $K_{\max}$ are derived as explicit, closed-form algebraic expressions directly from the physical corridor bounds and aerodynamic interval

widths, completely independent of numerical optimization logic. In the spherical-LVLH coordinates the corridor bounds are box constraints on (r, V) , so the interval widths that enter $C_{\mathcal{R}}$ and $K_{\max}$ have no wrapping from corridor intersection—the constants inherit the tightness of the coordinate choice. Requiring the constants in closed form, rather than as the output of a solver, is what separates a certificate from a computation: exact rational certificate representations are known to exist, with a documented bit-complexity that grows with the number of variables [29, 30]. Because interval arithmetic evaluates transcendentals natively, the traditional algebraic adjunctions required to force the system into a polynomial format [31] are obsolete. The state space carries only the physical dimensions—spherical position, wind-frame velocity, attitude quaternion, body rates, thermal, ablation, and modal states—with no artificial embeddings.

The ultimate guarantee relies on deriving the residual corridor constant $C_{\mathcal{R}}$ and the maximum boundary-layer bubble count $K_{\max}(Q_{\max})$. The number of bubbles is bounded a priori by the integrated heat-load budget. We must derive explicit, closed-form algebraic expressions for $C_{\mathcal{R}}$ and $K_{\max}$ strictly from the physical corridor bounds (which are box constraints on (r, V) in the spherical-LVLH state), the initial conditions, and the aerodynamic interval widths, entirely without relying on numerical interior-point solvers.

The shift to exact interval arithmetic renders traditional Savageau-style polynomial adjunctions—the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and the polynomial surrogate for wall temperature (Varma–Savageau–Voit recasting [31])—obsolete, as transcendentals are evaluated natively. These auxiliary states are eliminated entirely, collapsing the state space to its minimal physical dimensions and removing a source of wrapping. The coordinate system is settled: translational position in geocentric spherical (r, λ, ϕ) , planet-relative velocity in wind-frame components (V, γ, ψ) , and attitude as a unit quaternion. This choice makes the entry corridor a box in state space (tightening interval intersections) and keeps the equilibrium-glide slow manifold native. The trade-off—transcendental dynamics versus polynomial dynamics with nonlinear corridor intersection—is resolved in favor of the spherical formulation, where wrapping is distributed across time steps and controlled by step-size refinement rather than concentrated in corridor intersection.

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